EDUKRIPA

Science Mock Test - 2

Class - 10th (2025 - 26) Time : 1.5 Hrs Max. Marks : 40

Section - A

1. Analyze the following chemical equations and identify the correct values for 'p' and 'q'.

Equation 1 : $pMg(s) + O_2(g) \rightarrow 2MgO\ (s)$

Equation 2 : $2Pb(NO_3)_2\ (s) \rightarrow qPbO(s) + 4NO_2(g) + O_2\ (g)$

(a) p = 1, q = 1 (b) p = 2, q = 4 (c) p = 1, q = 2 (d) p = 2, q = 2

2. When dissolved in water, which of the following oxides is most likely to form a solution with a pH value greater than 7?

(a) SO_2 (Sulphur dioxide) (b) CO_2 (Carbon dioxide) (c) K_2O (Potassium oxide)

(d) N_2O_5 (Dinitrogen pentoxide)

3. A student places a clean strip of magnesium metal into a test tube containing a blue solution of copper(II) sulphate. What is the expected observation after some time?

(a) The solution remains blue, and a greyish layer deposits on the strip.

(b) A colourless, odourless gas is evolved, and the strip dissolves completely.

(c) The blue colour of the solution fades, and a reddish-brown solid is deposited on the strip.

(d) No reaction occurs, and the appearance of the solution and the strip remains unchanged.

4. Arrange the following common substances in the increasing order of their pH values: Lemon Juice, Toothpaste, Sodium Hydroxide Solution, Distilled Water.

(a) Lemon Juice < Toothpaste < Distilled Water < Sodium Hydroxide Solution

(b) Distilled Water < Lemon Juice < Toothpaste < Sodium Hydroxide Solution

(c) Lemon Juice < Distilled Water < Toothpaste < Sodium Hydroxide Solution

(d) Sodium Hydroxide Solution < Toothpaste < Distilled Water < Lemon Juice

A few drops of phenolphthalein indicator are added separately to a solution of vinegar and a solution of washing soda. What are the observed colours?

	Colour in Vinegar	**Colour in Washing Soda Solution**
A	Pink	Colourless
B	Colourless	Pink
C	Orange	Yellow
D	Colourless	Colourless

5.

6. Dilute sulphuric acid is added to sodium hydrogen carbonate. The gas evolved is then passed through freshly prepared lime water. What is the final observation?

(a) A reddish-brown gas is seen, which turns lime water orange.

(b) No gas is evolved.

(c) Brisk effervescence occurs, and the gas turns lime water milky.

(d) The solution becomes hot, and the lime water remains clear.

7. Answer the following based on the properties of metals:

(a) The electrical wiring in homes is typically made of copper or aluminium. Which property of these metals makes them suitable for this purpose?

(b) Gold can be drawn into extremely thin wires, some just a few micrometers thick. Name the property responsible for this characteristic.

8. Attempt either option A or B.

(A) The chief ore of a metal 'M' is Cinnabar (HgS). This metal is placed low in the reactivity series.

(i) Identify the metal 'M'.

(ii) Name the process used to convert the sulphide ore into its oxide and write the balanced chemical equation for the reaction.

(iii) How is metal 'M' subsequently extracted from its oxide? Write the equation.

OR

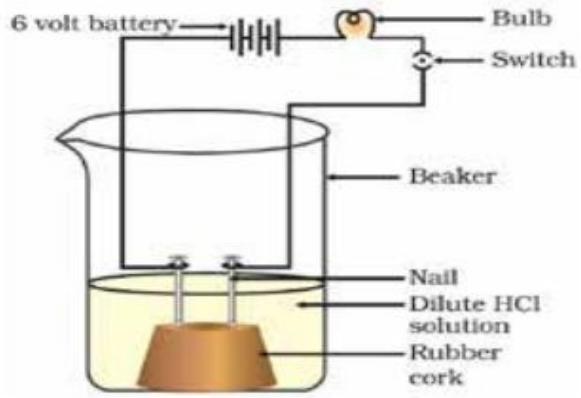
(B) Alloys are prepared to enhance the properties of metals.

(i) Name the constituent elements of the alloy Brass.

(ii) State two reasons why stainless steel, an alloy of iron, is preferred over pure iron for making cooking utensils.

(iii) What is an alloy containing mercury as one of its constituents called? Give an example.

A student performs an experiment to test the electrical conductivity of a dilute acid solution using a beaker, two graphite rods, a battery, and a bulb.



(a) When a dilute solution of hydrochloric acid (HCl) is taken in the beaker and the switch is turned on, what will the student observe? Give a reason for this observation.
(b) If the HCl solution is replaced with an aqueous solution of glucose ($C_6H_{12}O_6$), will the observation change? Explain why.
(c) Name a common base which, if used instead of HCl, would produce a similar observation as seen in part (A).

9.

10. While playing in a garden, Anjali was stung by a honeybee, which caused her immense pain and a burning sensation. Her friend immediately advised her to rub the affected area with some toothpaste. Later that evening, after having a spicy meal, her father complained of acidity. Anjali suggested he take a solution of baking soda.

(a) Bee stings are acidic in nature. How does rubbing toothpaste on the sting provide relief?

(b) Why did Anjali suggest baking soda for her father's acidity?

(c) Name the acid injected by the bee sting.

OR

A laboratory has three unlabelled beakers containing different liquids : (A) Rainwater with a pH of 5.2, (B) Tap water with a pH of 7.1, and (C) Drain cleaner with a pH of 13.5.

(a) Classify each liquid as a weak acid, a weak base, or a strong base.

(b) What is acid rain? Explain why liquid (A) is an example of acid rain.

(c) Which of the three liquids will have the highest concentration of hydrogen (H+) ions?

Section - B

11. A student is analyzing the behavior of light rays and makes the following observations:

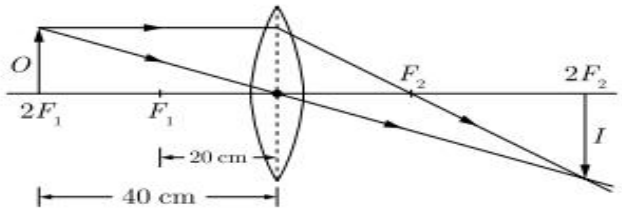
I. A ray of light passing through the center of curvature of a concave mirror retraces its path.

II. A ray of light parallel to the principal axis of a convex mirror, after reflection appears to diverge from the principal focus.

III. A ray of light incident obliquely at the pole of a concave mirror is reflected obliquely, making an equal angle with the principal axis.

Choose from the following the correct option that lists the correct statements.

(a) I and II (b) I and III (c) I, II and III (d) II and III



12. The above image shows the formation of an image with an optical instruments.
A. Which type of the lens is used in the above diagram ?
B. What type image is formed in the above case?
C. What is the focal length of the lens

13. Assertion (A) : A convex lens of focal length 20 cm is placed in front of an object at 10 cm. The image formed will be virtual and erect.

14. Reason (R) : For a convex lens, when the object is placed between the optical center and the principal focus, it forms a virtual and erect image.

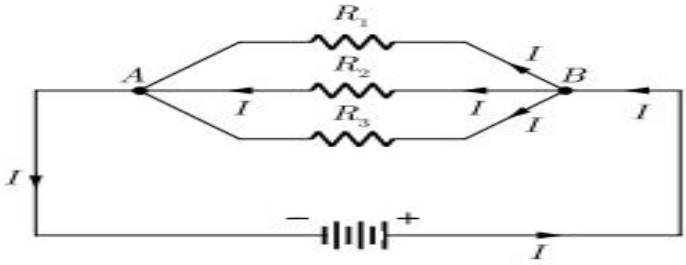
(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, and R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. In the given circuit diagram suppose the resistors R1, R2 and R3 have the values 5 Ω , 10 Ω , 30 Ω respectively, which have been connected to a battery of 12 volt. Calculate



(a) The current through each resistor,

(b) The total current in the circuit

15. (a) What is the total resistance of n resistors each of resistance R connected in : (i) Series, (ii) Parallel.

(b) Calculate the resultant resistance of 3 resistors 3 Ω , 4 Ω and 12 Ω connected in parallel.

16. Every car today is fitted with rear-view mirrors for safety purposes. The outside rear-view mirror is specially designed to give a wider field of view so that the driver can see vehicles coming from behind. These mirrors are made slightly curved outward and provide an image that looks smaller than the actual object, but helps cover a larger portion of the road. That is why the message "Objects in the mirror are closer than they appear" is written on such mirrors. This practical application is based on the principle of image formation by spherical mirrors.

A. Which type of spherical mirror is used here—concave or convex?

B. State two characteristics of the image formed in such a mirror for distant objects.

C. For a convex mirror of focal length f =− 30cm, a vehicle is u = − 500cm from the mirror. Find the

image distance v .

OR

D. For the situation in (C), find the magnification m and comment on the image size.